due to dissipative forces.

6 **(a)** When a p.d. of 240 V is applied across it, then the power dissipated by the lamp is 100 W. B1

(b) Resistance of lamp $R = \dfrac{V^2}{P} = \dfrac{240^2}{100} = 576\ \Omega$

Current flowing through the filament ,

$$I = \frac{V}{R} = \frac{280}{576} = 0.486\ \text{A}$$

In one second, quantity of charge flowing through it, Q = 0.500 C C1

$$\frac{n}{t} = \frac{Q}{e} = \frac{0.486}{1.6\times10^{-19}} = 3.04\times10^{18}\ \text{s}^{-1}$$

A1

(c) The temperature (or internal energy) of the filament rises with work done by the electrons on the filament. B1

This causes more vigorous lattice vibration of the atoms resulting in greater collision frequency between electrons and atoms, causing the resistance to rise. B1

To include explanation that just involve collisions of electrons with lattice ions.

7 **(a)** $A = A_0 e^{-\lambda t}$